

HW 10

1

For each incompletely specified flow table in (a), (b), (c), below determine:

1. the pair table,
2. the maximal compatibility classes,
3. a minimal closed covering (and explain why it is minimal),
4. a second minimal closed covering if the first is not unique, and
5. the corresponding reduced flow table.

a

✓ Answer ✓

(A, B)

not (A, C)

(A, D) if (B, C) and (C, E)

not (A, E)

(B, C) if (B, C) and (C, E)

(B, D) if (B, C) and (C, E)

(B, E)

not (C, D)

(C, E) if (C, E)

not (D, E)

(A, B)

(B, C, E)

(A, D)

Our minimal cover would be (A, D), (B, C, E) as it only requires two different groups to cover the entire function.

	X1	X2	X3
A	B, 0	B, 1	B, d
B	B, 0	B, 0	A, d

b

✓ Answer

	A	B	C	D	E	F
A	o	x	x	x	x	o

	A	B	C	D	E	F
B	-	0	x	x	x	x
C	-	-	0	x	x	x
D	-	-	-	0	x	x
E	-	-	-	-	0	x
F	-	-	-	-	-	0

(A, F), B, C, D, E

Its the only possible combination

	X0	X1
A	A, 0	A, 0
B	B, 0	C, 0
C	E, 0	A, 1
D	B, 0	D, 0
E	A, 1	D, 0

C

✓ Answer

	A	B	C	D	E	F
A	0	C, E	B, D C, F C, E C, D	x	C, F C, D D, E	x
B	-	0	x	C, E	x	x
C	-	-	0	D, E C, F C, D	0	C, D D, E C, F
D	-	-	-	0	C, F C, D D, E	x
E	-	-	-	-	0	x
F	-	-	-	-	-	0

(A, B), (C, E)

(B, D), (C, E)

	00	01	11	10
A	A, d	B, 0		C, d
B	C, d	D, 0	B, d	

	00	01	11	10
C	B, 0	B, 0		A, d
D		-, 1	C, d	A, d

	00	01	11	00
A	B, d	C, 0		B, d
B	C, 0	C, 0		A, d
C	B, d	D, 0	C, d	
D		-, 1	B, d	B, d

2

You will design a controller for a traffic light at a junction of a single-track railroad and a highway (Fig 1). Trains passing through the junction press and release switches. When a train approaches the junction from either direction and is within 1500 feet from it, the lights should change from green to red and remain red until the train is 1500 feet past the junction. Trains do not back up, and consecutive trains are always at least 3000 feet apart.

a

First assume that each train is shorter than 3000 feet in length. Formulate a primitive flow table with comments for each row; seven stable states suffice. Using a pair table, determine the maximal compatibility classes, find a minimal closed covering, and derive the corresponding reduced flow table.

✓ Answer

	0	X1	X2	Output
N	N	Left	Right	0
Right	Right	L2	Right	1
Left	Left	Left	R2	1
L2	N	L2	Right	1
R2	N	Left	R2	1

b

Repeat part (a), assuming that trains can be any length. Nine states suffice for the primitive flow table.

✓ Answer

	0	X1	X2	Output
N	N	Left	Right	0

	0	X1	X2	Output
Right	Right	L2	Right	1
Left	Left	Left	R2	1
L2	N	L2	Right	1
R2	N	Left	R2	1

3

A factory produces steel bars of length $L + D$ and $L - D$. You will design the controller of a digital system to segregate the bars (Fig 2.). Bars arrive on a conveyor belt and pass under two photocells P1 and P2 as shown below. The spacing between successive bars on the belt is greater than D . To the right of P2 is a trap door through which short bars drop. The door should not be open when the beam of P2 is interrupted and should be open immediately after a short bar of length $L - D$ has completely passed P2. For $i = 1, 2$ let signal x_i be 1 when the beam of P_i is interrupted. To open the trap door, set control signal z to 1.

	0	P1	P2	P12	z
N	(N)	O			0
O	S	(O)	L	L	0
S	(S)	S2	S'		0
L	N	O	L	(L)	0
S'	N	O	(S')	O1	0
O1	S	O'		(O1)	0
O'	S	(O')	S'	L	1

4

For each of the reduced flow tables below, find a state assignment with no critical races and a minimum of internal variables. Variables w and x are the inputs

a

A: 00
 B: 01
 C: 11
 C': 10 → 11

b

A: 00
 B: 01

D: 10

C $\rightarrow$ A is a concern, but C is stable so it is okay

C

AC

BA

BD

CD

CB

DB

DA

A: 00

B: 01

C: 11

D: 10

The only concern is $C \rightarrow A$, but since the target is stable, it is okay

d

AB

AD

BD

BE

CD

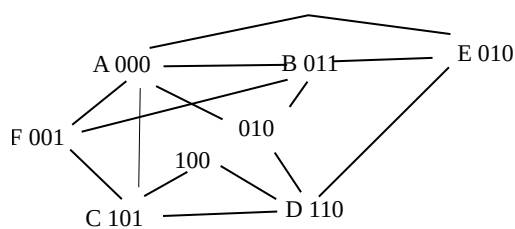
CF

DE

EA

FA

FB



A: 000

B: 011

C: 101

D: 110

E: 010

F: 001

010: DD--

100: -D

e

AB

AC

AD

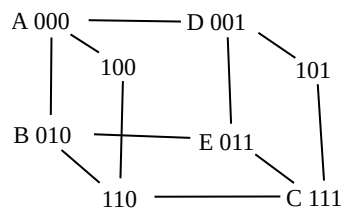
BC

BE

CD

CE

DE



A: 000

B: 010

C: 111

D: 001

E: 011

110: (100)B--

100: A---

101: --C-